higher education & training

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NATIONAL CERTIFICATE

LOGIC SYSTEMS N5

(8080295)

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09:00–12:00

This question paper consists of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
LOGIC SYSTEMS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL circuits must be fully labelled.
5. Waveforms should preferably be drawn with the clock input on top followed by the LSB flip-flop output below the clock input and with the MSB flip-flop output as the last output waveform at the bottom.
6. Write neatly and legibly.

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QUESTION 1

A printer requires a binary (digital) counter which has the following output states:

110_2. 101_2, 100_2, 011_2, 010_2 and 001_2.

Design a synchronous binary counter that produces the above output states.

The counter must be self-starting and must exit from any unused state to the starting state within ONE clock pulse.

Design and draw the circuit making use of a truthtable, Karnaugh maps and THREE JK master/slave flip-flops. **[15]**

QUESTION 2

A scanner requires a binary (digital) counter to generate the following output states:

010_2 011_2 100_2 101_2 110_2 and 111_2

Draw the truth table, circuit and the input and output waveforms for an asynchronous binary counter which uses gated reset/set to produce the above output states.

Use THREE JK master/slave flip-flops in the circuit.

The truth table and waveforms must clearly show the reset action. **[15]**

QUESTION 3

3.1 The counting of a signal's pulses for a fixed accurate period of time, usually one second, is known as frequency measurement.

Draw a fully labelled block diagram of a typical frequency counter. (7)

3.2 Draw a typical set of waveforms for the frequency counter in QUESTION 3.1. (5)

[12]

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QUESTION 4

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.1–4.15) in the ANSWER BOOK.

4.1 Four left shifts of 0100_2 in a ring counter results in 1000_2.

4.2 Two right shifts of 1000_2 results in 0010_2.

4.3 Two left shifts of 0001_2 results in 0100_2.

4.4 0001_2 and 1000_2 are both Johnson and ring code output states.

4.5 A ring counter and a Johnson counter are one and the same.

4.6 A Johnson code shift register has inverted feedback from its MSB flip-flop output to its LSB flip-flop input.

4.7 A synchronous counter can generate the Johnson code.

4.8 A parallel-in serial-out shift register converts serial input to parallel output.

4.9 The parallel output of a circulating shift register is connected to its serial input.

4.10 The serial input of a circulating shift register is connected to its serial input.

4.11 Series-in series-out shift registers are also known as S.I.S.O registers.

4.12 1000_2 is a Johnson code output state.

4.13 A shift of data from 0001_2 to 0100_2 is called right-shifting

4.14 Shift registers are used to count clock pulses.

4.15 Shift registers are used mainly to store and shift data.

(15 × 1) **[15]**

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QUESTION 5

Choose a description from COLUMN B that matches an item in COLUMN A. Write only the letter (A–O) next to the question number (5.1–5.15) in the ANSWER BOOK.

COLUMN A		COLUMN B	
5.1	5_{10} plus 7_{10} in XS3BCD is …	A	11011_2
5.2	5_{10} plus 7_{10} in 8421BCD is …	B	adds or subtracts 2 multibit numbers using 1's compliment
5.3	1100_2 plus 1001_2 plus 0110_2 is …	C	shift left and subtract process
5.4	1101_2 times 101_2 is …	D	adds two bits plus a carry-in
5.5	11101_2 minus 10100_2 is …	E	slower than the parallel ALU
5.6	The 2's compliment of negative 011010 is …	F	shift right and add process
5.7	The 1's compliment of negative 011010 is …	G	arithmetic logic unit
5.8	ALU stands for …	H	adds two multibit numbers
5.9	The full adder	I	adds or subtracts 2 multibit numbers using 2's compliment
5.10	The parallel adder	J	01001_2
5.11	The parallel 1's compliment ALU	K	100110_2
5.12	The parallel 2's compliment ALU	L	0100 0101
5.13	The multiplication process	M	100101_2
5.14	The division process	N	0001 0010
5.15	The series adder ALU	O	1000001_2

(15 × 1) **[15]**

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QUESTION 6

Computer memory is required to store the program instruction words and the data words. Each 1 or 0 in each program word or data word is stored in several memory cells for each word.

6.1 Define the term *static read and write memory* (SRAM). (2)

6.2 Define the term *volatile memory.* (2)

6.3 Define the term *dynamic read and write memory (DRAM)*. (2)

6.4 Define the term *non-volatile memory* (NVM) and give TWO examples of NVM. (4)

6.5 Draw a fully labelled diagram of a typical SRAM memory cell. (4)

[14]

QUESTION 7

Complete the following sentences by only writing the missing word next to the question number (7.1–7.14) in the ANSWER BOOK.

In a typical drum-printer design, a fixed font (7.1) … set is engraved onto several radial tracks along the length of a large drum (cylinder), which (7.2) … at high speed and paper and an inked ribbon are stepped (moved) past the print position. As the desired character for each column passes the print position, a (7.3) … strikes the paper from the rear and (7.4) … the paper against the ribbon and the drum, causing the desired (7.5) … to be recorded on the continuous paper.

A dot-matrix printer or impact-matrix printer is a type of computer printer with a (7.6) … head that runs back and forth, or in an up and down motion, on the page and prints by (7.7) … an ink-soaked cloth ribbon against the (7.8) …, much like the print mechanism of a typewriter. However, unlike a typewriter or daisywheel printer, letters are drawn out of a dot (7.9) …, and thus varied fonts and arbitrary graphics can be produced. Because the printing involves mechanical pressure, these printers can create carbon copies and carbonless copies.

Each dot is produced by a tiny metal (7.10) …, also called a 'wire' or 'pin', which is driven forward by the power of a tiny (7.11) … or solenoid, either directly or through small levers (pawls).

Chain printers (also known as train printers) place the type on moving bars, a horizontally-moving (7.12) … As with the drum printer, as the correct (7.13) ... passes by each column, a (7.14) … is fired from behind the paper. **[14]**

TOTAL: 100

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